

```
.data
DecryptStringSpace: .space 400 # 400 bytes of space, more than enough...
```

```
AVCorrect:      .asciiz "\nCorrect Assessment"
AVKeyName:      .asciiz "keyval "
AVFailLead:     .asciiz "Test "
AVFalsePos:     .asciiz ": failed with correct key\n"
AVFalseNeg:     .asciiz ": failed (says correct) with wrong key\n"
AVStartTest:    .asciiz "\nStarting Test "
NewLine:        .asciiz "\n"
newline:        .asciiz "\n"
AVDone:         .asciiz "\nALL DONE\n"
```

```
TestData:
# format is good key, bad key, encodephrase-len (in words), encode phrase
.word 12, 25, 2, 0x44414337, 0x39483940, 0
.word 17, 30, 6, 0x4334312f, 0x2f413443, 0x3d303743, 0x3534312f, 0x2f344
13e, 0
.word 0, 0, 0
```

```
.text
```

```
main:
```

```
    la $t0, TestData
    li $t1, 0
    li $t4, 1
```

```
AddVerifyTestLoop:
```

```
    add $t2, $t0, $t1
    lw $t3, 0($t2)
    beq $t3, $zero, AddVerifyTestDone
    la $a0, AVStartTest
    li $v0, 4
    syscall
    move $a0, $t4
    li $v0, 1
    syscall
    la $a0, NewLine
    li $v0, 4
    syscall
```

```
    move $a2, $t3
    sll $a2, $a2, 8
    or $a2, $a2, $t3
    sll $a2, $a2, 8
    or $a2, $a2, $t3
    sll $a2, $a2, 8
    or $a2, $a2, $t3
    move $a0, $t2
    add $a0, $a0, 12
    la $a1, DecryptStringSpace
    sw $t0, -4($sp)
    sw $t1, -8($sp)
    sw $a0, -12($sp)
    sw $a1, -16($sp)
    sw $a2, -20($sp)
    sw $t2, -24($sp)
    sw $t4, -28($sp)
```

```

addi $sp, $sp, -28
jal AddAndVerify
addi $sp, $sp, 28
lw $t0, -4($sp)
lw $t1, -8($sp)
lw $a0, -12($sp)
lw $a1, -16($sp)
lw $a2, -20($sp)
lw $t2, -24($sp)
lw $t4, -28($sp)
li $t3, 1
beq $t3, $v0, AddVerifyNextCase

```

```

la $a0, AVFailLead
li $v0, 4
syscall
move $a0, $t4
li $v0, 1
syscall
la $a0, AVKeyName
li $v0, 4
syscall
lw $a0, 0($t2)
li $v0, 1
syscall
la $a0, AVFalsePos
li $v0, 4
syscall

```

Now try with bad key

AddVerifyNextCase:

```

lw $t3, 4($t2)
move $a2, $t3
sll $a2, $a2, 8
or $a2, $a2, $t3
sll $a2, $a2, 8
or $a2, $a2, $t3
sll $a2, $a2, 8
or $a2, $a2, $t3
move $a0, $t2
add $a0, $a0, 12
la $a1, DecryptStringSpace
sw $t0, -4($sp)
sw $t1, -8($sp)
sw $a0, -12($sp)
sw $a1, -16($sp)
sw $a2, -20($sp)
sw $t2, -24($sp)
sw $t4, -28($sp)

```

```

addi $sp, $sp, -28
jal AddAndVerify
addi $sp, $sp, 28
lw $t0, -4($sp)
lw $t1, -8($sp)
lw $a0, -12($sp)
lw $a1, -16($sp)
lw $a2, -20($sp)
lw $t2, -24($sp)
lw $t4, -28($sp)

```

```
beq $zero, $v0, AddVerifyNextIter
```

```
la $a0, AVFailLead
li $v0, 4
syscall
move $a0, $t4
li $v0, 1
syscall
la $a0, AVKeyName
li $v0, 4
syscall
lw $a0, 4($t2)
li $v0, 1
syscall
la $a0, AVFalseNeg
li $v0, 4
syscall
```

```
AddVerifyNextIter:
```

```
la $a0, DecryptStringSpace
li $v0, 4
syscall
addi $t4, $t4, 1
lw $t3, 8($t2)
sll $t3, $t3, 2
add $t1, $t1, $t3
addi $t1, $t1, 16
j AddVerifyTestLoop
```

```
AddVerifyTestDone:
```

```
la $a0, AVDone
li $v0, 4
syscall
li $v0, 10
syscall
```

```
##### Put Code for AddAndVerify, IsCandidate, WordDecrypt Here
```